

Width, Memory, and Delay: A Resource Accounting for the Limits of Flat Multi-Agent Systems

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Abstract—A recurring question in the design of scalable multi-agent systems—from robot swarms to collectives of large-language-model (LLM) agents—is whether adding more agents can, on its own, overcome performance limits, or whether a qualitatively *deeper* organization is required. A recent preprint argues that flat, homogeneous multi-agent systems face an irreducible, population-independent “causal floor” on achievable error, removable only by hierarchical (nested-loop) organization. Using a controlled disturbance-rejection testbed with an exactly computable optimum, we show this conclusion is too strong and replace it with a quantitative resource model built on three resources: population *width* N , per-agent internal-model *memory* d , and prediction across the observation *delay* τ . We establish three claims. (i) The achievable floor is governed not by architectural hierarchy but by per-agent internal-model content: a flat, homogeneous swarm whose agents carry a matched internal model of the disturbance matches or beats a designed two-loop hierarchy at equal per-agent memory—so temporal depth can be dynamical (recurrent memory), not architectural (nesting). (ii) The three resources are *not mutually interchangeable*; we chart the exchange rates and the hard non-exchange boundaries on an explicit width $\times$ memory map, including a strict equal-total-state-budget comparison. (iii) A residual floor is set by the observation delay and the environment’s unpredictability over that horizon, which we verify against the optimal controller (the exact minimum-variance characterization is given in Proposition 2). We quantify the price of replacing oracle knowledge of the disturbance spectrum with online learning, provide a preliminary robustness check against a mild bounded nonlinearity and a spatially-extended plant, and distill four design rules for practitioners. All experiments are released with a reproducible pipeline.

Index Terms—Multi-agent systems, scalability, minimum-variance control, internal model principle, disturbance rejection, time delay, adaptive control, resource trade-offs, large-language-model agents.

I. INTRODUCTION

A PRACTICAL question now dominates the engineering of multi-agent systems: does adding more agents keep paying off? In the language-model-agent literature the debate is explicit and unresolved—several recent studies report that

a well-tuned single agent can match or beat elaborate multi-agent workflows [1], [2], while others find that scaling helps only when the added agents bring genuine *diversity* rather than replication [3]. The same tension appears in robot swarms and networked control, where consensus and averaging arguments promise gains with scale [4]–[6], yet some limits appear stubbornly independent of population size.

A recent preprint [7] sharpens the pessimistic side into an impossibility claim: flat, homogeneous multi-agent systems possess a population-independent “causal floor” on achievable error, and only hierarchical (nested-loop) organization can lower it. The intuition is attractive—some limits do seem to demand a deeper temporal organization—but, we argue, the claim conflates two distinct notions under the word “flat”: *flat in architecture* (no nested control loops) and *flat in memory* (agents without internal dynamical state). The impossibility result holds for the second notion and fails for the first, and the two come apart exactly where the interesting engineering lives.

This paper replaces the single flat/deep axis with a *resource triangle*—width, memory, and delay-prediction (Fig. 1)—and asks, for each pair of resources, whether one can be traded for another and at what rate. The answer is a mix of clean exchange laws and hard prohibitions, all recoverable from classical stochastic control theory (minimum-variance control [8], [9], the internal model principle [10], and the Bode/waterbed constraints [11], [12]) once the problem is posed as resource accounting rather than architecture. Our contribution is not a new control law; it is the reframing, an explicit map of where exchange is and is not possible, and a careful empirical verification—against a computable optimum—that pins the constants. Because the accounting is stated in terms of averageable noise, structured disturbance, and predictability across a delay, it is agnostic to whether an “agent” is a controller, a robot, or an LLM; we return to this bridge in the conclusion.

Contributions

- 1) A counterexample to the strong flat-systems theorem: a flat homogeneous swarm with matched per-agent internal models beats a designed hierarchy (Section VI, Figs. 2 and 4).
- 2) A quantitative width $\times$ memory map with exchange rates and non-exchange boundaries, plus a fixed-budget ($N \cdot d$ const) analysis (Figs. 4 and 5).
- 3) A delay theorem, stated in resource terms and verified to $< 1\%$ against the optimal predictive controller (Fig. 10).

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The scalable-control trilemma

three resources, three distinct jobs — none substitutes for another

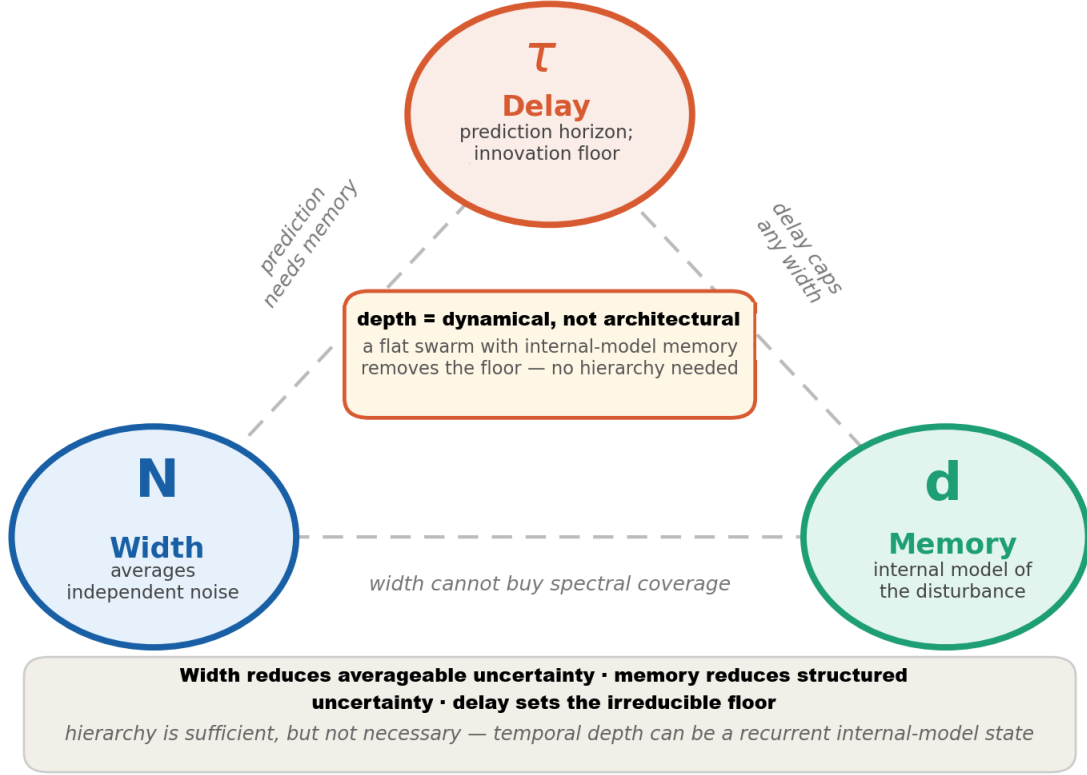


Fig. 1. The resource triangle. Width averages independent measurement noise; per-agent internal-model memory covers the structured disturbance; delay sets an irreducible innovation floor. A memoryless flat swarm cannot lower the structured floor, but a flat swarm whose agents carry an internal model can—so temporal depth may be dynamical (memory), not architectural (nesting).

- 4) The cost of adaptation: how an oracle-free learned internal model approaches the oracle, and where it fails (Figs. 7 and 8).
- 5) Four practitioner rules and a nonlinear-plant robustness check (Section VII, Fig. 11), plus a spatially-extended plant confirming the accounting is not an artifact of the scalar testbed (Fig. 12).

II. RELATED WORK

Minimum-variance and self-tuning control. The floor we characterize is the minimum-variance benchmark of Åström [8]: for a linear plant with delay and a disturbance with rational spectrum, the minimum achievable output variance equals the variance of the multi-step-ahead prediction error of the disturbance, obtained by polynomial division of the noise model. The self-tuning regulator [9] identifies that model online; Harris [13] turned the benchmark into a closed-loop performance monitor still used industrially. Our adaptive baseline is in this lineage in spirit—it identifies the disturbance online and cancels it—but is not a full AR/RLS minimum-variance self-tuning regulator; it is an adaptive internal-model controller specialized to narrowband disturbances (periodogram- based

frequency identification driving a bank of learned resonators, Section III). We are explicit about this to avoid claiming more than the implementation delivers.

Internal model principle. Francis and Wonham [10] showed that asymptotic rejection of a disturbance requires the controller to embed a model of that disturbance’s generator. This is exactly the “per-agent memory buys spectral coverage” axis: a resonator is the internal model of a sinusoid, an integrator that of a constant bias, an AR model that of a colored process.

Fundamental limitations. Bode’s sensitivity integral and its delay-constrained descendants—the “waterbed effect” [11], [12]—formalize that suppression pushed down in one band re-emerges elsewhere and that delay caps the bandwidth over which suppression is possible. We use these to explain the notch-depth/robustness-width trade-off (Fig. 7) and the delay wall (Fig. 10) rather than re-deriving them. The Smith predictor [14] and Kalman filter [15] supply the optimal predictive machinery for the delay analysis.

Multi-agent scaling and emergence. Consensus and cooperative control establish $\propto 1/N$ averaging of independent measurement noise [4]–[6]. The LLM-agent literature is now testing whether analogous scaling holds for cognitive agents,

with mixed conclusions [1]–[3], [16]. The target preprint [7] sits at this intersection; we treat its claim as the null hypothesis to be refined.

Distributed estimation and graph-theoretic internal models. We deliberately study two extremes: fully decentralized agents with no communication (our width axis) and a fully centralized Kalman fusion (our frontier benchmark, Section III). Between them lies a large literature this paper does not engage experimentally: distributed/consensus Kalman filtering lets agents partially fuse measurements over a communication graph without a fusion center [17], and the internal model principle has been extended to networked, graph-constrained settings via cooperative output regulation, where each agent embeds a copy of the exogenous signal generator and consensus over the graph substitutes for centralized coordination [18]. Limited communication could in principle let width partially substitute for memory by letting agents share spectral estimates; our resource accounting does not yet include this regime, and we flag it as the most direct extension of this work (Section VIII).

III. PROBLEM SETUP AND TESTBED

From here through Section VII the setting is a physical control system; the disturbance is a stochastic signal with known statistics, not a model of cognition. The bridge to LLM-agent collectives—where “memory” is context and retrieval and “delay” is reasoning latency—is a motivating analogy, taken up only in the introduction, the discussion (Section IX), and Appendix A; it is not part of the control-theoretic claims.

Plant. Our testbed is deliberately minimal: a *single controlled scalar object* observed by N homogeneous agents whose control actions are averaged and whose measurement noises are independent. Concretely, one state x evolves under a common disturbance and each agent i contributes a control u_i from its own noisy, delayed observation:

$$x[t+1] = x[t] + \bar{u}[t] + d[t] (+\gamma \sin x[t]), \quad \bar{u}[t] = \frac{1}{N} \sum_i u_i[t], \quad (1)$$

where the optional term introduces plant nonlinearity. This is not a swarm of N independent plants but the canonical setting in which *width* (more agents) and *memory* (per-agent internal state) can be varied orthogonally; Section VIII discusses the spatially-extended generalization. The common disturbance is

$$d[t] = \sum_b A_b \sin(\phi_b[t]) + p[t], \quad p[t] = a p[t-1] + e[t], \quad (2)$$

with ϕ_b accumulating at frequency ω_b (permitting chirp and regime switching) and $e[t]$ white. Each agent observes

$$y_i[t] = x[t-\tau] + v_i[t], \quad v_i \sim \mathcal{N}(0, \sigma_m^2), \quad (3)$$

with v_i *independent across agents*—this is what population width can average away; the common disturbance is what it cannot. *The delay is in the observation, not in the actuation:* $\bar{u}[t]$ enters $x[t+1]$ immediately, but each agent must choose $u_i[t]$ from information no fresher than $x[t-\tau]$. The control problem is therefore predictive—the controller must forecast the state and disturbance over a horizon of $\tau+1$ steps from the delayed, noisy observation.

Aggregate view for centralized baselines. For the mean of N agents with i.i.d. measurement noise, $\bar{y}[t] = x[t-\tau] + v[t]/\sqrt{N}$, so width enters the strong baselines *only* as measurement-noise reduction—a clean separation of the width axis from the memory axis. The Kalman frontier is a *centralized* reference built on this ensemble mean: it is the best achievable performance *if the agents’ measurements could be perfectly fused*. Decentralized agents with no communication cannot attain it—each sees only its own noisy y_i —so we use it purely as an upper-bound benchmark and a computable optimum for the resource accounting, not as a controller the swarm could implement.

Metric. Steady-state MSE = $\mathbb{E}[x^2]$ after burn-in, averaged over seeds with 95% confidence intervals. We also fit $\text{MSE}(N) = AN^{-\alpha} + C$ and report the floor C with bootstrap CIs.

Controllers.

- *Ladder (per-agent memory d)*: a homogeneous flat swarm whose agents carry an internal model of increasing order—P ($d=0$), PI ($d=1$), and PI+resonator banks (PIRES, $d = 1 + 2k$ for k resonators). The integrator covers bias and very-low-frequency drift; each resonator adds one narrowband component, so memory depth is natural and monotone.
- *Cascade*: a two-loop hierarchy (the preprint’s “deep” system).
- *Kalman frontier*: oracle-model Kalman-predictive minimum-variance control; the computable optimum, with a semi-analytic Riccati floor.
- *Adaptive*: oracle-free adaptive internal-model control—identifies the disturbance tones online (peak-picking on an exponentially-averaged periodogram of a decimated, pre-filtered reconstruction) and cancels them with phase-locked-loop learned resonators plus soft feedback. It is a *heuristic engineering baseline*, not a self-tuning minimum-variance regulator: it uses periodogram peak-picking rather than recursive least squares and has no formal convergence guarantee. We include it to quantify the gap between oracle frequency knowledge and practical online learning. It also carries additional estimator memory (a spectral buffer, oscillator bank, and filter states) beyond $2k+1$ states, so we report it as a no-oracle baseline and exclude it from equal-memory comparisons.

IV. FRAMEWORK: THREE RESOURCES

We define the resources so the exchange questions are well posed.

Width N . The number of agents sharing the common disturbance and averaging independent measurement noise; the resource that reduces the *averageable* error component.

Per-agent memory d . The dimension of each agent’s internal dynamical state, updated as $m_i[t+1] = f(m_i[t], y_i[t])$. We distinguish *internal-model* memory (state encoding a generator of the disturbance: integrator, resonator, AR model) from generic recurrence; only internal-model memory buys spectral coverage.

Temporal depth / prediction. The organization of memory across separated timescales, used to predict the plant state

and disturbance over the delay $\tau + 1$ from the delayed, noisy observation. This is what the preprint intuitively as “depth”; we argue it is not architectural nesting but the presence, in the loop, of a recurrent internal model whose prediction horizon spans τ . When that model is fixed we call it *recurrent internal-model memory*; when it also updates its model online and uses it to predict across the delay we call it *adaptive predictive memory*. (We avoid “recursive”: in a control/RNN context “recurrent” is accurate.)

The refined thesis, in one line: *Population width reduces only averageable uncertainty. Internal-model memory reduces structured uncertainty. Temporal depth is not architectural hierarchy but predictive closed-loop state organized across the relevant timescales. Delay sets the irreducible innovation floor.*

V. PROPOSITIONS

We state the results as propositions with proof sketches; empirical verification follows in Section VI.

Assumption 1: (A1) linear plant; (A2) disturbance with rational spectrum (finite sum of narrowband tones plus an AR process); (A3) per-agent i.i.d. measurement noise, independent across agents; (A4) known plant delay τ . Under (A1)–(A4) the minimum-variance machinery of [8] applies to the aggregate.

Proposition 1 (Width–structure non-exchange): Consider controllers of a fixed class and bounded bandwidth, and let the disturbance contain power P_b in a frequency band whose generator is not embedded in any agent’s internal model. Then the steady-state error contributed by that band is bounded below by a positive constant independent of N ; increasing N cannot reduce it. (Width lowers the *averageable* noise component but not the structured residual, whose *asymptotic* rejection requires an internal model of the band; a broadband robust controller may partially attenuate the band but cannot drive its contribution to zero.)

Proof sketch: Width acts only on the per-agent i.i.d. term $v_i/\sqrt{N}$ in the aggregate; the common disturbance $d[t]$ is identical across agents and survives averaging unchanged. Rejecting power in band b requires loop gain shaped at ω_b , which by the internal model principle [10] requires an internal model of that band in the loop. Absent such a model, the closed-loop sensitivity at ω_b is $\Theta(1)$ in N , so the band’s contribution to $\mathbb{E}[x^2]$ is bounded below by a constant. Verified in Fig. 3 and by the L-shaped contours of Fig. 4. ■

Proposition 2 (Memory–delay partial exchange; the delay floor): Under Assumption 1, the minimum achievable steady-state variance equals the conditional variance of the controlled next state given the delayed, noisy observation filtration:

$$C_{\min}(\tau) = \text{Var}(x[t+1] - \hat{x}[t+1 | \mathcal{F}_t^y]), \quad (4)$$

$\mathcal{F}_t^y = \sigma(y[s], u[s-1] : s \leq t)$, $y[s] = x[s-\tau] + v[s]$. This has two contributions: (a) the innovation of the disturbance accumulated over the horizon $\tau + 1$, which no controller can anticipate, and (b) the residual state-estimation uncertainty from observing only the delayed, noise-corrupted $x[t-\tau]$. Under perfect state measurement it reduces to the classical minimum-variance floor—the variance of the $(\tau+1)$ -step disturbance prediction error [8]—but with noisy delayed sensing the estimation term must be included.

Proof sketch: Because control acts through the plant without its own delay, $x[t+1]$ is fixed by $x[t]$, the applied $u[t]$, and $d[t]$; the best mean-square $u[t]$ cancels $\mathbb{E}[x[t] + d[t] | \mathcal{F}_t^y]$, leaving $\text{Var}(x[t+1] | \mathcal{F}_t^y)$. Writing $d = (C/A)e$ with e white and applying the $(\tau+1)$ -step division $C/A = F + z^{-(\tau+1)}G/A$ isolates the unpredictable innovation Fe ; the delayed noisy observation adds the Kalman estimation covariance of the current state, which the Riccati recursion of the optimal filter [15] yields in closed form. The sum is $C_{\min}(\tau)$; it grows with τ , with the disturbance memory (larger f_k for more strongly colored spectra), and with the sensing noise σ_m^2/N . Verified to $< 1\%$ in Fig. 10. ■

Corollary 1 (Delay wall): No amount of width or memory removes $\text{Var}(Fe)$; it is the environment’s unpredictability over the latency horizon.

Proposition 3 (Depth without hierarchy): There exists a flat, homogeneous swarm (identical agents, no nested loops) whose per-agent internal model matches the disturbance and whose steady-state floor is at or below that of a designed two-loop hierarchy with the same per-agent memory. Hence architectural hierarchy is sufficient but not necessary to lower the floor.

Proof sketch: Constructive. Give each agent a PIREs internal model tuned to the disturbance bands (memory $d = 1 + 2 \cdot \#\text{bands}$). With the bands cancelled, the flat swarm’s floor is the delay/innovation floor of Proposition 2. It suffices to exhibit one flat controller that beats one designed hierarchy: our PIREs swarm attains a lower floor than the implemented two-loop cascade at equal per-agent memory (Fig. 2), so architectural hierarchy is not necessary in general. (We do not claim the flat controller beats *every* hierarchy—a better-tuned or differently-organized cascade might match it; the impossibility claim we refute is universal, so a single counterexample is enough.) Verified in Fig. 2 and the $d=7$ row of Fig. 4. ■

Proposition 4 (Cost of adaptation, empirical): The heuristic no-oracle controller of Section III approaches a tuned internal-model (oracle-frequency) baseline in stationary narrowband settings, with a gap that is controlled empirically by the online identification error and shrinks as the per-population SNR (hence N) grows. Below an identification threshold N_{id} its floor is uncontrolled, and under regime switches faster than the identification time it is dominated by re-identification transients and may exceed that of a robust memoryless loop. We state this as an empirical observation, not a convergence theorem: our controller is an engineering baseline, not a self-tuning minimum-variance regulator, and a formal convergence analysis is beyond this paper’s scope. A tighter regulator (e.g. RLS) would likely narrow the stationary gap; the transient failure under fast switching, however, is set by the re-identification time and is therefore fundamental to any learning scheme, not an artifact of our estimator.

Justification: For the classical AR/RLS self-tuning regulator the attained limit is the minimum-variance floor [9]; our resonator-bank controller only shares the same qualitative behavior for narrowband disturbances, without a convergence guarantee. Identifiability requires the disturbance to rise above the reconstruction noise, which scales as σ_m^2/N ; below that the periodogram peaks are buried. Under switching faster than

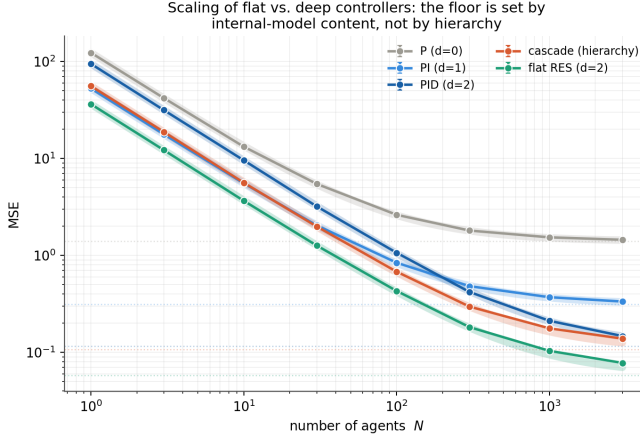


Fig. 2. E1–E3: $MSE(N)$ with fitted floors. A flat resonator swarm ($d=2$) attains a lower floor than the two-loop cascade at equal per-agent memory, refuting the strong flat-systems claim by counterexample.

the estimator’s memory the estimate never settles. Observed in the adaptive row of Fig. 4 and in Figs. 7–8. ■

VI. EXPERIMENTS

Phase 1 (E1–E5) establishes the dichotomy and the counterexample; Phase 2 (Blocks A–E) turns it into the quantitative accounting and adds the optimal frontier and the oracle-free adaptive baseline. Two exploratory studies (E6, E7) are deferred to Appendix A.

E1–E3 — counterexample (Fig. 2). Floors (95% CI) on a single-band task: P 1.40, PI 0.32, PID 0.12, CASCADE 0.107, flat RES 0.058. The flat resonator swarm ($d=2$) beats the two-loop cascade at equal per-agent memory. To make the comparison fair to the hierarchy, the cascade’s loop gains were selected by a grid search (48 combinations over its proportional, slow-loop-bandwidth, and slow-loop-integral gains) minimizing MSE on the tuning task; the best-tuned gains are then frozen and reused across all N , exactly as for every other controller. The flat swarm still attains the lower floor. All scaling exponents $\alpha \approx 1.0$ (pure $1/N$ noise reduction to a fixed floor). *Supports Prop. 3.*

E4 — width cannot buy coverage (Fig. 3). On a two-band task, an internal model covering one band leaves the other’s power N -independent (MSE at $N=1000$: 2.96 uncovered vs. 0.25 both covered). *Supports Prop. 1.*

E5 — latency waterbed. PI MSE rises $0.61 \rightarrow 3.29$ as $\tau: 1 \rightarrow 10$ and destabilizes by $\tau=20$; the PSD shows the waterbed. *Motivates Prop. 2.*

Block A — the width $\times$ memory map (Figs. 4, 5). MSE over $N \times d$ on a 3-band task. Floors at $N=1000$: $d=0$ 4.49, $d=1$ 2.40, $d=3$ 2.05, $d=5$ 1.66, $d=7$ 0.44 (monotone, since the PIRES ladder adds an integrator then one resonator per band); no-oracle adaptive baseline 0.47. The heatmap contours are L-shaped: width slides you down a noise ramp to a fixed floor; memory lowers the floor itself. A strict equal-budget comparison (exact $N \cdot d = B$ points, Fig. 5) shows a threshold: at the smallest budget ($B=1050$) going deep is roughly a wash ($d=7$ with $N=150$ gives 2.24 vs. 2.37 for $d=1$ with

$N=1050$), because too few agents leave the measurement noise unaveraged; but past a minimum SNR ($B \geq 3150$) spending the same states on memory dominates decisively ($d=7$: 0.80 at $B=3150$, 0.29 at $B=12600$, vs. ≈ 2.3 for $d=1$ at every budget). This directly answers the “you just gave each agent more memory” objection. *Quantifies Props. 1 and 3.* The mechanism behind the ladder is visible in the residual spectrum (Fig. 6): width scales the broadband noise floor, while each unit of internal-model memory notches out one more disturbance band.

Block B — robustness and the oracle’s price (Figs. 7, 9, 8). *Mismatch $\times$ damping*: a sharp resonator ($\rho=0.9999$) reaches $MSE \approx 0.22$ within $\pm 10\%$ of ω but degrades to ≈ 0.64 at 30% detune—worse than model-free PI (0.49); a soft resonator ($\rho=0.99$) is ≈ 1.0 everywhere. Depth of suppression $\leftrightarrow$ width of robustness is a conserved trade-off (a Bode waterbed inside the notch), which the full mismatch $\times$ damping surface (Fig. 9) confirms is continuous across the whole parameter plane, not just at the three damping levels plotted in Fig. 7. The adaptive controller (0.42) beats the *strongly* detuned oracles ($\geq 30\%$ error) and the soft resonator, but a *well-centred* sharp resonator still wins where it is accurate ($MSE \approx 0.22$ – 0.33 within $\pm 15\%$). *Nonstationary*: under chirp the adaptive controller wins (0.40 vs. 0.41 oracle, 0.51 PI); under regime switching PI (0.59) edges out adaptive (0.79) because of re-identification transients—reported honestly. *Supports Prop. 4 and its failure mode.*

Block C — the delay theorem (Fig. 10). Across 15 (a, τ) combinations, the simulated optimal MSE matches the semi-analytic Riccati floor of Prop. 2 to $< 1\%$; PI diverges past $\tau \approx 10$ – 15 . The floor grows with τ , the noise memory a , and the sensing noise. *Verifies Prop. 2.*

Block E — a mild-nonlinearity robustness check (Fig. 11). As a preliminary check—not a claim about general nonlinear plants—we add a bounded term $0.15 \sin x$ (this is neither chaotic nor non-minimum-phase). The memory ordering holds *once width is sufficient* ($N=300$: $d=2$ $1.76 < d=1$ $2.32 < d=0$ 3.65), but at small $N=10$ integrative memory can amplify measurement noise—PI ($d=1$, 16.4) is then worse than plain P ($d=0$, 13.6), while the resonator ($d=2$, 6.5) still leads. Under nonlinear+switching the stale oracle resonator (2.07) and adaptive (2.45) both beat PI (3.75). The linear resource accounting is thus robust to this class of *bounded* nonlinearities; generalization to unbounded nonlinearities (e.g. x^3 , or non-minimum-phase plants) is left for future work.

Block F — spatial generalization (Fig. 12). To check that the accounting is not an artifact of the scalar testbed, we replace the single object with a ring of 12 diffusively-coupled integrator nodes driven by a *shared slow mode* plus local per-node noise; each node runs a local controller fed by N agents’ delayed, noisy observations. Once the population is large enough to average the local noise, the same ordering emerges: the memoryless controller (P) saturates at a floor (≈ 3.7) independent of N because the shared structured mode survives averaging, the integrator (PI) does better, and a resonator tuned to the shared mode reaches the lowest floor (0.23 at $N=300$). Width cures the local noise; only memory cures the shared structure—exactly as in the scalar case. This

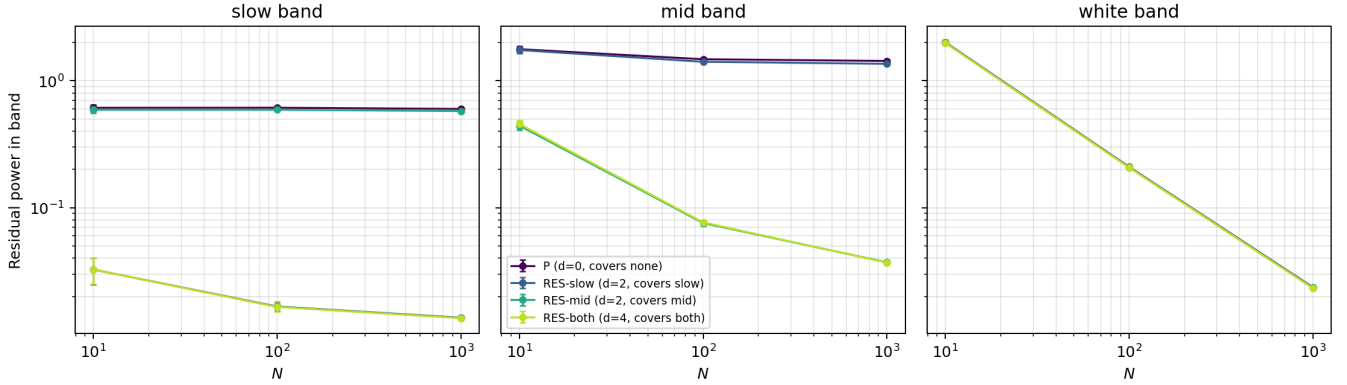
E4: residual power per spectral band — uncovered bands do not improve with N 

Fig. 3. E4: an internal model covering one band drives its power down with N ; the uncovered band's power is N -independent. Width cannot buy spectral coverage.

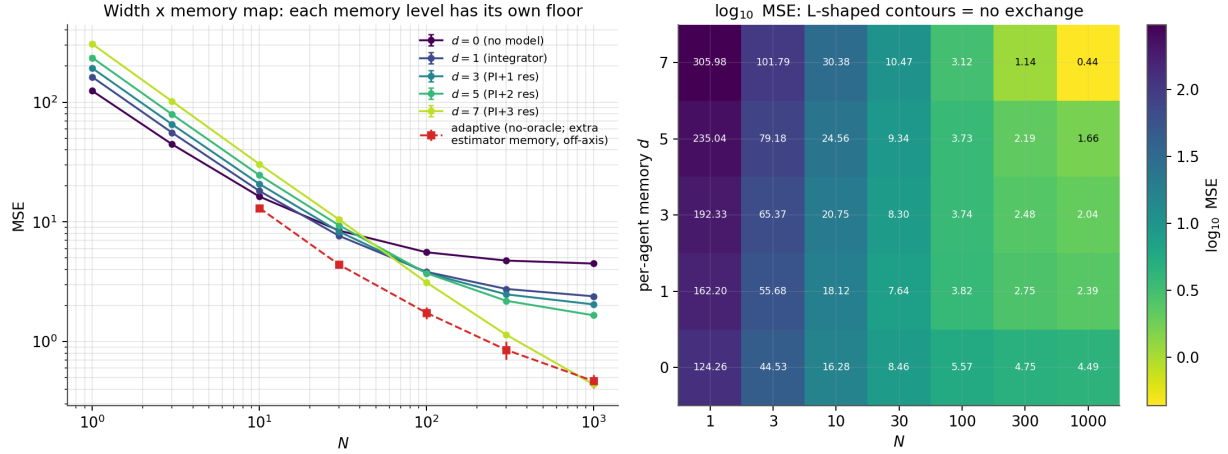


Fig. 4. Block A: MSE over $N \times d$ on a 3-band disturbance. Left: each per-agent memory level has its own floor; the adaptive controller (red) starts worse at small N and approaches the oracle $d=7$ floor at large N . Right: heatmap colored by $\log_{10}$ MSE (annotated numbers are the raw MSE, not its log); the L-shaped contours mean width and memory are not interchangeable—width slides down a noise ramp to a fixed floor, while only memory (covering another band) lowers the floor itself.

further holds on a two-dimensional 6×6 grid and, more strikingly, under *partial observability* (Fig. 13): when only half the nodes carry sensors and actuators, width alone (P) is stuck at a large floor (≈ 18.5), but a resonator on the shared mode still attains the lowest error—the memory ordering survives both the coupling dimension and incomplete sensing.

VII. THE RESOURCE TRIANGLE AND DESIGN RULES

Table I summarizes the accounting, and Fig. 14 reads it off the width $\times$ memory map as three regimes: a variance-limited corner cured by width, a memory-limited corner cured by internal-model states, and the delay/innovation floor that neither removes.

Golden rules.

- 1) *Count the disturbance modes first.* Each persistent band needs its own internal-model state (~ 2 states/band). No number of agents substitutes for a missing mode.

- 2) *Add width only up to N^* .* Beyond the point where $1/N$ reduction meets the structural floor, agents are ballast; N^* is computable from σ_m^2 and the target floor.
- 3) *Latency is the hard wall.* The floor equals the environment's unpredictability over the observation (feedback) delay. Buy a predictor or cut latency, not more agents.
- 4) *Under model uncertainty, detune or adapt.* Trade notch depth for robustness width; go adaptive only if regimes outlive the identification transient.

VIII. LIMITATIONS AND OUTLOOK

The core testbed is a single scalar object observed by N homogeneous agents, which isolates the width/memory/delay axes; Block F confirms the accounting survives on a spatially-extended chain and a two-dimensional grid, including under 50% partial observability (Figs. 12, 13). Further, the plant is a 1-D integrator per node (a nonlinear variant is tested, but genuinely chaotic, higher-order, or non-minimum-phase plants are

TABLE I
THE RESOURCE TRIANGLE: WHAT EACH RESOURCE BUYS, WHAT IT CANNOT, AND THE GOVERNING LAW.

Resource	Buys	Does <i>not</i> buy	Rate / law
Width N	averaging of i.i.d. noise	structural/spectral content	$\propto 1/N$ down to a floor; saturates at N^*
Per-agent memory d	spectral coverage, internal model	delay compensation by itself	need $d \geq 2 \cdot \# \text{bands}$; floor set by uncovered power (Prop. 1)
Prediction over τ	cancels the <i>predictable</i> part of the disturbance (shifts the floor curve down)	the <i>innovation</i> : the unpredictable noise over the horizon is never removed	floor = disturbance innovation over $\tau+1$ plus delayed-sensing estimation uncertainty (Prop. 2)
Adaptation	replaces oracle knowledge of ω	instant tracking	$\approx$ oracle — small gap; loses under fast switching (Prop. 4)
Resonator damping	width of robustness	depth at that width	notch waterbed: depth $\times$ bandwidth $\approx$ const

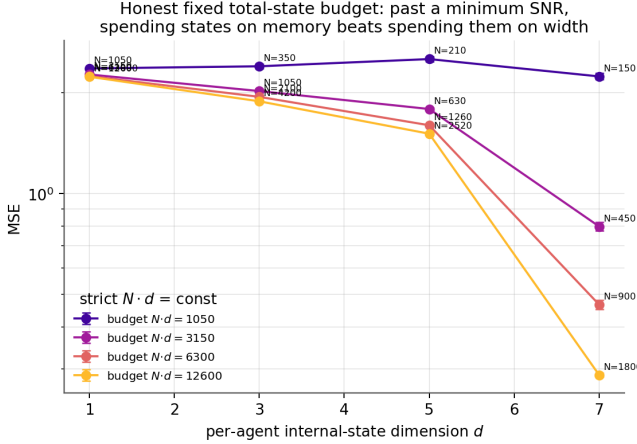


Fig. 5. Strict equal-budget comparison ($N \cdot d = B$, exact points, budgets multiples of $\text{lcm}(1, 3, 5, 7)$). At the smallest budget ($B=1050$) measurement noise dominates, so width and depth are equivalent; above a threshold SNR, depth (internal-model memory) dominates. This answers the “you just gave each agent more memory” objection.

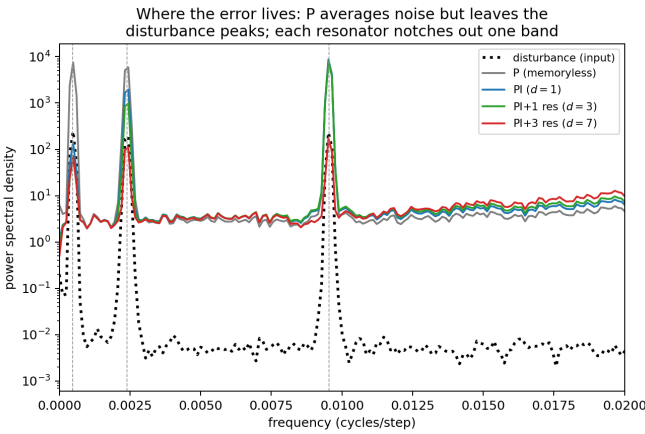


Fig. 6. Mechanism, in the spectrum. The memoryless P controller averages noise but leaves the disturbance peaks intact; the integrator suppresses the lowest band; each resonator notches out one more band. Memory reshapes the residual spectrum, width only scales it—the internal-model principle made visible.

not); the tuning grid is coarse (the *ordering* of floors is robust, though constants may shift); the adaptive controller assumes

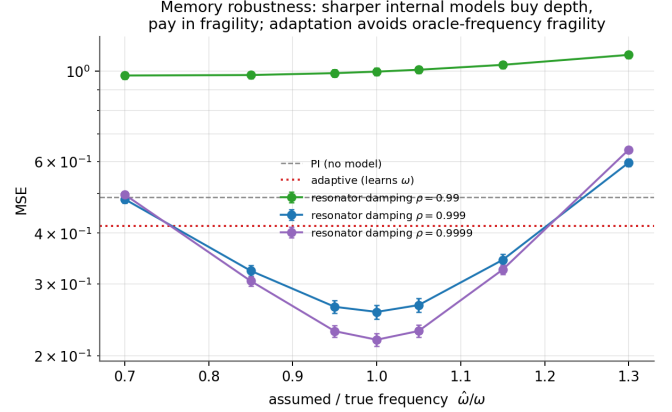


Fig. 7. Block B: frequency mismatch $\times$ resonator damping. Sharper internal models buy suppression depth but pay in fragility; adaptation re-centres the notch and escapes the strongly-mistuned regime.

a few narrow tones (broadband or heavy-tailed disturbances are out of scope); it loses to a robust PI under fast switching (reported, not tuned away). The hierarchical baseline is a single tuned two-loop cascade; Proposition 3 is a counterexample to a universal claim and does not rule out a better-designed hierarchy (e.g. H_∞ or model-based nested predictors) matching the flat swarm. We also assume no inter-agent communication; Section III discusses how distributed/consensus estimation could partially close the gap to the centralized frontier. Measurement noise is i.i.d. across agents and actuators are unconstrained; correlated or heterogeneous sensing and actuator saturation are not modeled. We do not claim to prove impossibility results; we exhibit a counterexample to a strong claim and propose a sharper, classically-grounded resource model, verified empirically. Each of these—a second hierarchical design, distributed estimation, richer plant classes, correlated/heterogeneous noise, and input constraints—is a concrete, testable extension of the same resource accounting rather than a departure from it.

IX. DISCUSSION

A trilemma, not a dichotomy. The debate the target preprint joins is usually posed as flat *versus* deep. Our results recast it as a triangle of three non-interchangeable resources (Fig. 1): width buys averaging, internal-model memory buys

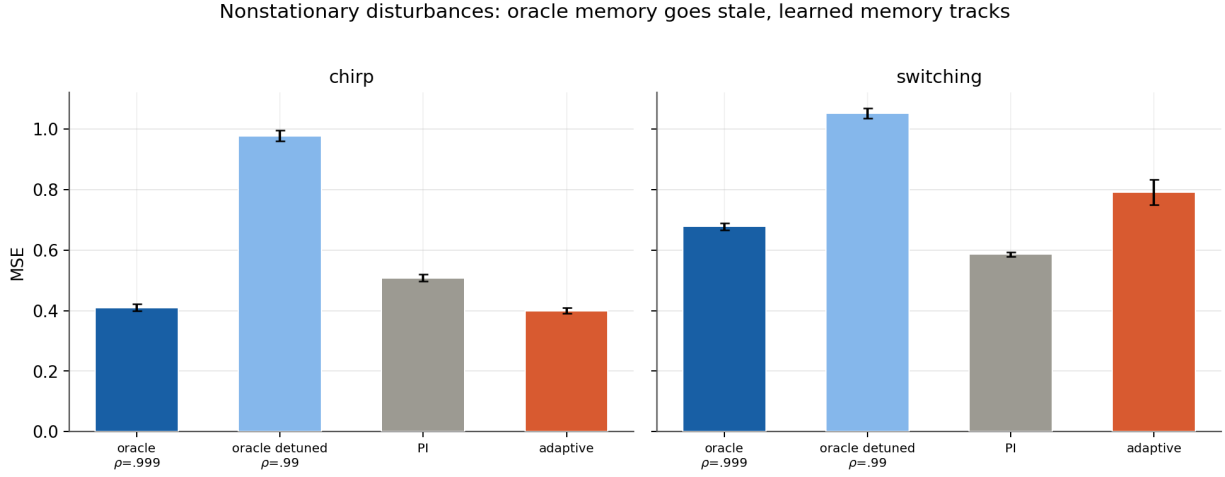


Fig. 8. Block B, nonstationary disturbances. Under a chirp, learned memory tracks; under regime switching, a robust memoryless loop (PI) can beat the adaptive controller because each switch triggers a re-identification transient.

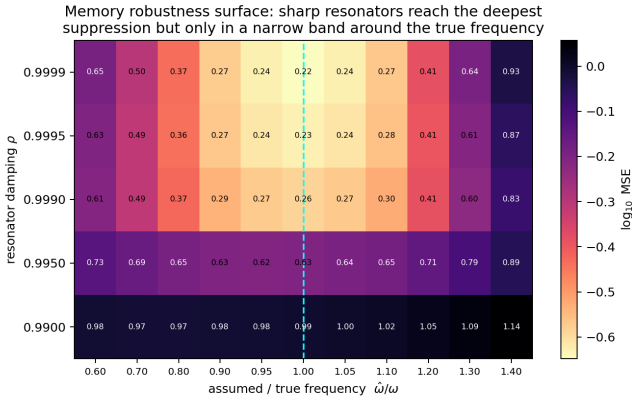


Fig. 9. Memory robustness surface (Block B). Color: $\log_{10}$ MSE over the two internal-model parameters, assumed/true frequency $\hat{\omega}/\omega$ and resonator damping ρ (annotated numbers are the raw MSE). Sharp resonators (top rows) reach the deepest suppression but only in a narrow band around the true frequency; damped resonators (bottom) are robust but shallow—the depth/robustness waterbed made quantitative.

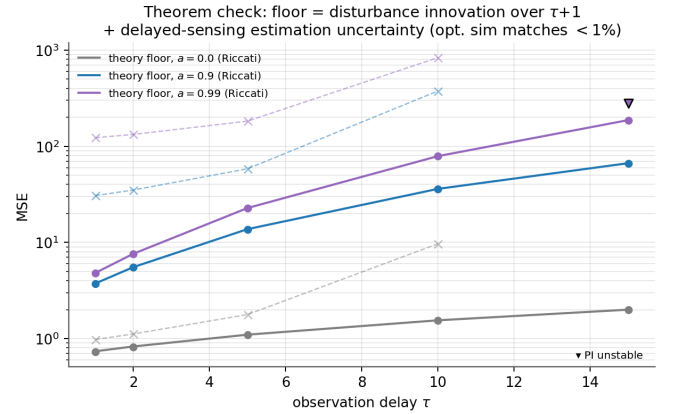


Fig. 10. Block C: the optimal controller’s simulated MSE (markers) matches the semi-analytic Riccati floor (lines) to $< 1\%$ (max 0.89%) across delays τ and noise memories a ; the floor is the disturbance innovation over $\tau+1$ plus the delayed-sensing estimation uncertainty (opt. sim matches $< 1\%$). PI (dashed) is shown where stable and marked ($\blacktriangledown$) where it diverges.

structure, and delay imposes a floor. Unlike the blockchain “scalability trilemma”, where one must sacrifice one of three desirable properties, here the three resources are not competing goods but *different jobs*: the lesson is that no amount of one substitutes for another, and that the “depth” the pessimistic view demands is bought by recurrent internal-model memory rather than by architectural nesting.

A biological reading. The account has a suggestive parallel in social insects. A colony does not build a hierarchy of “generals” to steer each ant; hierarchy is expensive and slow. Instead evolution hard-wires into each individual an internal model of the world—instinct—while the size of the swarm (width) averages out individual sensing noise. On our axes, instinct is per-agent internal-model memory, colony size is width, and the collective thrives precisely because structured regularities of the environment are encoded in every agent rather than computed by a central authority. The same account

predicts the failure mode: when the environment changes faster than instincts can be updated—faster than the identification time in our Block B switching experiment—the swarm cannot re-tune every individual quickly, and a rigid internal model is worse than a simple robust reflex. Two caveats keep the analogy honest: many insects retain limited online plasticity (bees learn), and evolution itself is a slow *outer* adaptation loop on a generational timescale—so the colony does adapt, but on a separated, much slower timescale, which is itself an instance of the depth-as-timescale-separation idea rather than a counterexample to it. We offer this as intuition, not mechanism.

Implications for LLM-agent systems. With the caveat that our testbed is a physical control system, the accounting offers a hypothesis for cognitive collectives. If “memory” maps to an agent’s context and retrieval (its internal model of the task domain), “width” to the number of parallel agents, and “delay” to reasoning or communication latency, then the

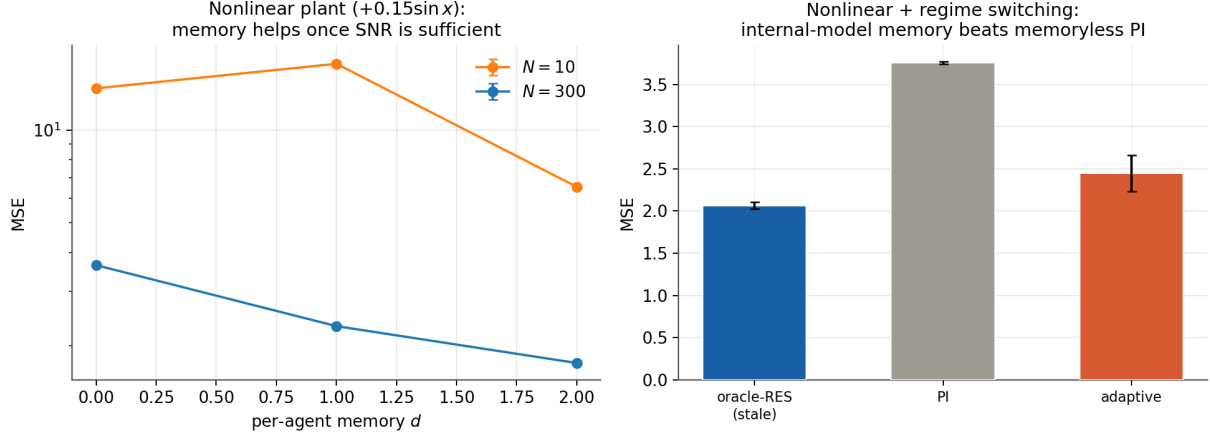


Fig. 11. Block E: on a nonlinear plant, the memory ordering persists once width is sufficient (left); under nonlinear plus regime switching, internal-model memory (a stale oracle resonator and the adaptive controller) beats memoryless PI (right). The oracle resonator still edges the adaptive controller here.

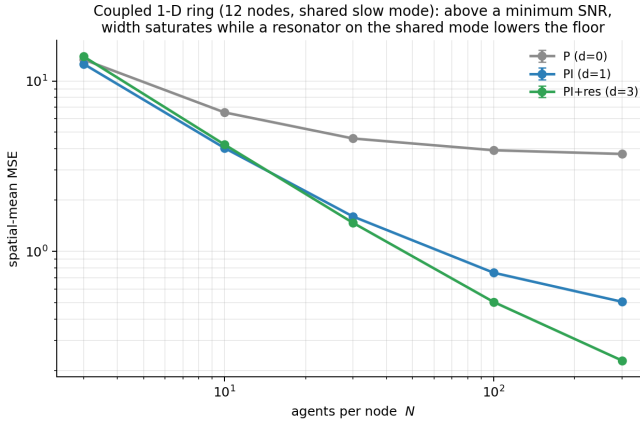


Fig. 12. Block F: a ring of 12 diffusively-coupled nodes with a shared slow disturbance mode. Width (P , gray) saturates at an N -independent floor because the shared mode survives averaging; an integrator (PI) improves; a resonator on the shared mode (green) reaches the lowest floor once the population is large enough to average the local noise (the ordering emerges above a minimum SNR). The scalar-case accounting carries over to the spatially-extended plant.

recurring empirical findings—that a strong single agent with adequate memory can match a multi-agent workflow [1], [2], and that scaling helps mainly when the added agents bring *diversity* [3]—are exactly what the trilemma predicts: replication (width) averages noise but cannot add missing domain structure, while diversity widens coverage. Whether the quantitative laws transfer is an open question our released harness is meant to probe.

X. CONCLUSION

Recasting the flat/deep question as resource accounting resolves the apparent paradox in both control swarms and LLM-agent collectives: width buys only the averageable part of the uncertainty, internal-model memory buys the structured part, and delay imposes an innovation floor that neither can remove. “Hierarchy” is one way to organize predictive memory

across timescales, not the only one—which is why a strong *single* agent with adequate memory can match a multi-agent workflow [1], [2], and why scaling helps mainly when it adds *diversity* that widens spectral coverage [3]. Whether the same accounting quantitatively governs language-model agents—where “memory” is context and retrieval and “delay” is reasoning latency—is a concrete and, we believe, testable next question; our released harness (Appendix A) is a first step.

APPENDIX A EXPLORATORY STUDIES

These studies are exploratory and not part of the main evidential line.

A. Emergent timescale specialization (E6). Flat collectives of learnable leaky integrators trained by an evolution strategy [19] reliably split off a *single* slow agent (learned timescales 20–1500 vs. 1–8 for the rest); the heterogeneous multi-band condition is more stable (0/5 unstable runs) than the homogeneous one (2/5). This suggests the slow state that Prop. 3 inserts by hand can instead *emerge* in a learned flat collective—heterogeneity creating a form of virtual temporal depth. We stress the exploratory status: populations are tiny (8 agents), the ES is simple, and a slow specialist appears (less consistently) even in the single-band control, so the evidence supports the heterogeneous-vs-homogeneous contrast, not a sharp multi-band claim.

B. LLM-agent analogue (E7, harness only). We provide a harness casting flat, flat-ensemble, and “deep” (fast-reactor plus slow-strategist) LLM agents onto the same regulation task, with a mock backend for pipeline validation and an API backend for real runs. The mock backend yields *no* scientific result; the harness is offered as a future benchmark for whether the width/memory/delay accounting transfers to language-model agents.

C. Adaptive controller dynamics (Fig. 15). The no-oracle adaptive controller is not a black box: its periodogram-driven frequency estimate converges to the true ω online, and its

2-D grid (6×6 coupled nodes, shared slow mode): the memory ordering survives coupling dimension and partial observability

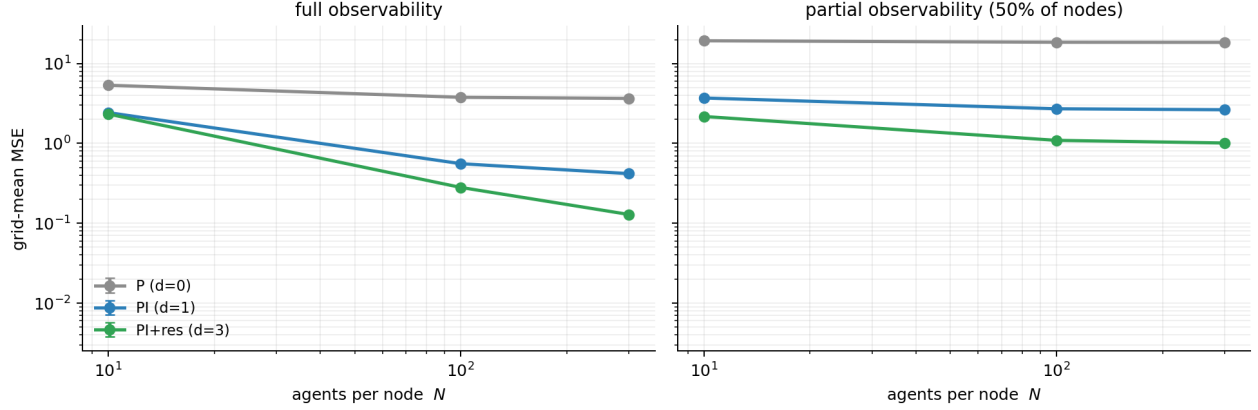


Fig. 13. Block F, continued: a two-dimensional 6×6 coupled grid. Left: full observability. Right: partial observability, where only half the nodes carry sensors/actuators and the rest are steered through diffusion. In both cases the memory ordering ($P > PI > PI+resonator$) is preserved; partial sensing raises every floor but does not change which resource lowers it.

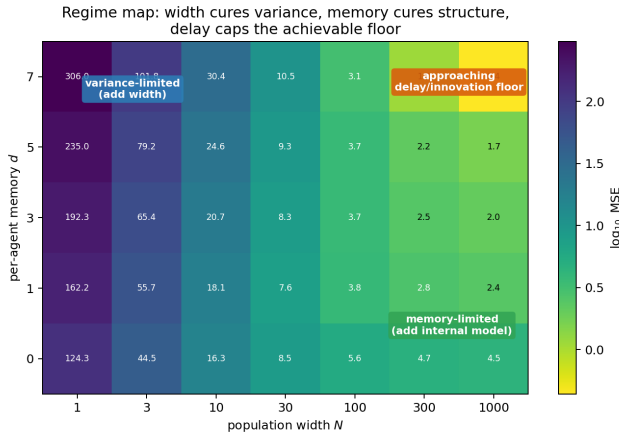


Fig. 14. Regime map, colored by $\log_{10} MSE$ (annotated numbers are the raw MSE). At low N the system is variance-limited (add width); at high N , low d it is memory-limited (add an internal-model state); the bottom-right corner approaches the delay/innovation floor that neither resource can lower.

sliding-window error approaches the oracle floor once the estimate locks, making concrete the threshold and gap of Proposition 4.

DATA AND CODE AVAILABILITY

All code, configurations, seeds, and raw results needed to reproduce every figure and number in this paper are released under the MIT license at <https://github.com/KuznetsovKarazin/causal-depth-limits>, with a versioned, citable snapshot archived on Zenodo (DOI: 10.5281/zenodo.21295164). Results regenerate deterministically from the released seeds; a Dockerfile, a Conda environment, and continuous integration (make smoke, make check) are included. No proprietary data or paid API access is required for the core results; the optional real-LLM harness (Appendix, make e7-real)

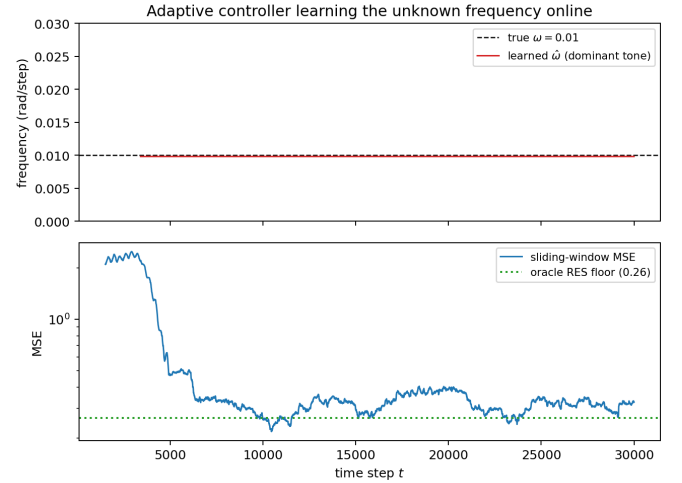


Fig. 15. Adaptive controller internals (no oracle). Top: the online frequency estimate locks onto the true unknown ω within a few thousand steps. Bottom: the sliding-window MSE falls toward the oracle resonator floor as the model converges—illustrating the identification threshold and convergence gap of Proposition 4.

needs an Anthropic API key and its output is excluded from the paper's claims.

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